

KRITHICK BALAJI RAMESH

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PROFESSIONAL SUMMARY

Embedded Software engineer with a B.Tech in Computer Science (GPA 8.44/10) and 1.5+ years building engineering applications across the full software life-cycle — from requirements to deployment and operations. Experience in **C/C++, Python, and real-time embedded targets**, with hands-on work in **modeling and simulation, electronics systems, and reliability-driven firmware**.

EDUCATION

SRM Institute of Science & Technology

B.Tech in Computer Science & Engineering — GPA: 8.44/10, First Class with Distinction

Chennai, India

Sept 2021 – Aug 2025

EXPERIENCE

GEETHIK Technologies – AI Research Intern

6 Months

India

Mar 2026 – Aug 2026

Dr. T. Vetrivel Chelian — Research Supervisor; R&D Specialist, Nokia, Bengaluru

Email: vetrivel.chelian_t@nokia.com

- Designed, developed, and delivered a production document-AI engineering application in Python across the full life-cycle (requirements, prototyping, iteration with stakeholders, deployment, maintenance).
- Applied model-based assurance and static analysis techniques with LLMs (Claude via the Anthropic API, GPT, BERT/T5) for layout-aware reconstruction, classification, and defect detection on noisy inputs — model-based assurance and LLM-driven verification
- Architected a modular system (classifier, detector, extractor, filter, renderer) enabling per-stage benchmarking and rapid research-to-production iteration on large corpora.
- Combined classical computer vision (deduplication, DPI normalization, layout filtering) with deep-learning detectors for scalable, quality-controlled extraction.

Wipro – Project Engineer — Trainee

4 Months

Chennai, India

Nov 2025 – Feb 2026

- Completed structured training in software engineering, Agile Scrum, and milestone-driven delivery; built a Java capstone applying systems-design principles.

SRM Institute of Science & Technology – Student Researcher

9 Months

Chennai, India

Mar 2024 – Nov 2024

Dr. Vijayalakshmi M. — Research Supervisor; Assistant Professor,

Department of Computing Technologies, SRM University

Email: vijayalm@srmist.edu.in — [Project Demo](#)

- Designed and delivered a real-time safety-monitoring system for power/gas hazard detection with automated fault response — spanning subsystem design, verification, and integration & test.
- Architected interrupt-driven firmware in Embedded C with deterministic control logic on ARM Cortex microcontrollers over SPI/I2C/UART, demonstrating analog-electronics and power-subsystem depth relevant to spacecraft power systems.
- Implemented multi-sensor data fusion (metal-oxide gas sensors) and cloud-synchronized alerting via RESTful APIs — reliability engineering and requirements verification in practice.

- Built a multimodal AI assistant (Gemini, BERT, T5, GPT) with NLP pipelines in Python on Azure AI Services for real-time inference.
- Optimized a code-similarity algorithm, reducing complexity from $O(n^2)$ to $O(n \log n)$.

TECHNICAL SKILLS

Languages	C, C++, Python.
OS / Real-Time	RTOS (FreeRTOS), ISR/NVIC, interrupt-driven firmware, ADC, SPI/I2C/UART.
Software Engineering	IDEs (VS Code), compilers, static analyzers, model-based assurance, independent verification, Git/GitHub/GitLab, JIRA, Agile Scrum, REST APIs.
AI	Algorithm & code development, LLM integration (Claude/Anthropic API, GPT, BERT, T5), NLP pipelines, ML deployment, edge/on-device inference.
Electronics & Signals	Power & sensor subsystems, ARM Cortex-M, ESP32, Raspberry Pi, GNU Radio, QPSK/BPSK/OFDM telemetry.

PUBLICATIONS

Full list: [Google Scholar](#) | [Research Profile](#)

Ramesh K. B., Saran S. — “Drone-Based Aerial Surveillance and Real-Time Hazardous Gas Leakage Detection System with Power BI Dashboard Integration.” *IEEE Xplore*, Jan 2025. [Link](#)

Ramesh K. B., Saran S. — “Multimodal AI-Enhanced Educational Assistant With Real-Time Q&A and Dynamic Learning Support.” *KUEY Journal*, vol. 31, no. 2, pp. 213–220, Jul 2025. [Link](#)

Ramesh K. B., Sivarankan P. G. — “Smart IntelliSecure-Based LPG and Fire Safety System with Integrated Sensor Fusion and Automated Actuation.” *Computing, Communication and Intelligence* (Routledge), Nov 2025. [Link](#)

Balaji K., Sivarankan P. G., Vijayalakshmi M. — “IoT-Enabled LPG Leakage Monitoring with Arduino, ESP32, and Real-Time Mobile Alerts via Flutter.” *SSRN*, Nov 2024. [Link](#)

AWARDS & DISTINCTIONS

Winner — SRM Project Expo 2025: Special Prize, Pico-SAT Mini Satellite Project.

2nd Place — SRM Project Expo 2024: Among 71 teams, LPG Leakage Detection System.

2nd Place — RAEEUCCI 2024 International Conference: Project Expo, ECE Department.

Knowledge Exchanger Award: KIXP, SRM Institute of Science & Technology.